

INSTITUTO SUPERIOR DE ENGENHARIA DO PORTO

Pós-Graduação em Sistemas Computacionais Embebidos
Sistemas Operativos de Tempo-Real (PGSCE-SIOPTR)

SIOPTR: PL1: Week 2

Scheduling and Synchronization

Paulo Baltarejo Sousa
pbs@isep.ipp.pt
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1 Real-time Systems

1. Consider an independent and periodic task set, executed in a pre-emptive context:

	C (ms)	T (ms)	D (ms)	Prio
τ_1	2	6	5	
τ_2	3	12	12	
τ_3	4	15	15	

- Assume a fixed priority assignment, based on Rate Monotonic. Assign priorities accordingly (fill in the table above): 3 (higher), 2 and 1 (lower). Justify your answer
 - For the previous task set, represent the time diagram for the RM scheduling, considering the time interval that validates the schedulability of the task set
2. Consider the following taskset:

	C (ms)	T = D (ms)	Arrival instant
τ_1	2	6	2
τ_2	3	12	1
τ_3	5	15	0

- Is this schedulable under preemptive EDF?
 - Trace the execution of the taskset in the first 20 ms
3. Consider the following taskset:

	C (ms)	T = D (ms)
τ_1	2	5
τ_2	1	9
τ_3	4	12

- Is this schedulable under non-preemptive EDF?
- Trace the execution of the taskset in the first 20 ms

4. Consider the following taskset:

	C (ms)	T = D (ms)
τ_1	9	10
τ_2	6	9
τ_3	4	7
τ_4	3	6

- Using Global EDF, trace the execution of the following task set with 3 processors for 15 ms
- Using Partitioned EDF, trace the execution of the following task set with 3 processors for 15 ms

5. Consider the following taskset:

	C (ms)	T = D (ms)	Arrival instant	Execution Sequence
τ_1	5	10	3	EERRR
τ_2	4	12	2	EEEE
τ_3	5	15	0	ERRRE

The execution sequence of each task is represented by two symbols(each one takes 1 ms to execute):

- The **E** represents free execution (without any constraint)
- The **R** represents a critical section managed by a resource sharing policy.

Trace the execution of the taskset in the first 13 ms, considering as resource sharing policy:

- None
- Priority Inheritance Protocol (PIP)
- Immediate Ceiling Priority Protocol (ICPP)

2 Operating System

1. Recurrent Task

- Download `ex1.tar.gz` from moodle SIOPTR webpage.
- Extract it, by:
> `tar -xvzf ex1.tar.gz`
- Change to `ex1` directory:
> `cd ex1`
- Compile it by:
> `make`
- Execute it by:
> `./task 1 5 500000000 10000000000 0`
- For analyzing it, prompt:
> `sudo trace-cmd record -e sched_switch ./task 1 5 500000000 10000000000 0`
- Then
> `kernelshark`

2. A set of recurrent task

- Download `ex2.tar.gz` from moodle SIOPTR webpage.
- Extract it, by:
`> tar -xvzf ex2.tar.gz`
- Change to `ex2` directory:
`> cd ex2`
- Compile it by:
`> make`
- Execute it by:
`> ./launcher`
- For analyzing it, prompt:
`> sudo trace-cmd record -e sched_switch ./launcher`
- Then
`> kernelshark`

3. A set of recurrent task (with semaphore)

- Download `ex3.tar.gz` from moodle SIOPTR webpage.
- Extract it, by:
`> tar -xvzf ex3.tar.gz`
- Change to `ex3` directory:
`> cd ex3`
- Compile and execute it by:
`> make run`

4. A set of recurrent Soft Real-Time tasks (with semaphore)

- Download `ex4.tar.gz` from moodle SIOPTR webpage.

- Extract it, by:
`> tar -xvzf ex4.tar.gz`
- Change to `ex4` directory:
`> cd ex4`
- Compile and execute it by:
`> make run`

5. A set of recurrent Hard Real-Time tasks (with semaphore)

- Download `ex5.tar.gz` from moodle SIOPTR webpage.
- Extract it, by:
`> tar -xvzf ex5.tar.gz`
- Change to `ex5` directory:
`> cd ex5`
- Compile and execute it by:
`> make run`